THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA FORM THREE PRE-MIDTERM EXAMINATION

032 CHEMISTRY 1
MARKING SCHEME AUGUST 2024

SECTION A (16 marks)

1. Multiple choice (10marks @ 01mark)

QNS	i	ii	iii	iv	v	vi	vii	viii	ix	x
ANS	C	B	C	A	D	E	A	E	B	B

2. Match the items (06marks @ 01mark)

LIST A	i	ii	iii	iv	v	vi
LIST B	B	F	C	H	E	A

SECTION B (54 marks)

3. (a)
   - Doors open outward for easy departure during emergency and for reducing the chance of knocking down laboratory apparatus and equipment.
   - The fume chambers are designed to capture and remove hazardous airborne substances generated during laboratory experiments that produce hazardous waste products are performed. (1½ @ = 03 marks)

(b) The following are important laboratory safety measures to obey:

(i) Label and lock all storage areas.

(ii) Be familiar with the location, use and limitations of the safety devices. This includes fire extinguishers, fire blankets, fume hood, spill cleanup materials, first aid kit, eyewash stations and fire alarm.

(iii) Keep all chemicals in properly labelled containers.

(iv) Be familiar with the appropriate safety measures to take when exposed to different hazardous materials.

(v) All chemicals that react with each other must be stored separately.

(vi) Use fume hoods/cupboards/chambers whenever possible.

(vii) Never store food in a refrigerator or freezer where hazardous chemicals are stored.

(viii) Stored chemicals must be inspected regularly to ensure they have not expired.

(ix) Keep the laboratory floor clean and dry at all times. Clean spills of water or chemicals immediately. (Any six:1 @ = 06 marks)

(x) All experimenters and other persons working in the laboratory should wear protective gears to minimize exposure to hazards. These gears may include lab coats, hand gloves, gumboots, safety goggles, aprons, etc.

4. (a) Solution: One amendment is that, the atom, which was considered to be indivisible during Dalton's time, was discovered to be composed of still smaller sub-atomic particles. …..03 marks

Another amendment is that atoms of the same element may have different masses. This is due to the presence of isotopes. Isotopes are atoms of the same element which have different mass numbers.

(03 marks)

(b) Solution

Percent abundance of ^{10}B × ^{10}B + Percent abundance of ^{11}B × ^{11}B = periodic table atomic mass

$\frac{20 \times 10.01}{100} + \frac{80 \times 11.01}{100} = 10.81$ (01 mark) (01 mark)

(01 mark)

5. (a) (i)

Solution

Chemical reactions are processes in which one or more substances are converted into other substances, while chemical equations are representations of the chemical reactions using the chemical formula of the reactants and products. ……02 marks

(ii) . Formation of Calcium carbonate, which… ……..……02 marks

Equations:

$CO_{2(aq)} + Ca(OH)_{2(aq)} \rightarrow CaCO_{3(aq)} + H_2O_{(l)}$ ……...01 mark

$CO_{2(aq)} + Ca^{+2}{}_{(aq)} + 2OH^-_{(aq)} \rightarrow Ca^{+2}{}_{(aq)}\ CO_3^{-2}{}_{(aq)} + H_2O_{(l)}$ ……..…...01 mark

$CO_{2(aq)} + 2OH^-_{(aq)} \rightarrow CO_3^{-2}{}_{(aq)} + H_2O_{(l)}$ ……..……...01 mark

(b)

(i) Synthesis ……..……..……..00½ mark

(ii) Decomposition ……...00½ mark

(iii) Precipitation ……..……..…..00½ mark

(iv) Redox ……00½ mark

6. (a)

K = magnesium (00½ mark) ; 2:8:2 I (00½ mark)

N = Fluorine (00½ mark); 2:7(00½ mark)

P = Argon (00½ mark); 2:8:8 (00½ mark)

L = Potassium (00½ mark); 2:8:8:1(00½ mark)

(b) Electrovalent (ionic),bond L_2Q ……..……01@=2 marks

Chemical properties:

- It conducts electricity when dissolved in water or in molten state .....01½ marks
- It has high melting and boiling points .....01½ marks

7. (a) Hard water is the water that does not readily form lather with soap/ water with high mineral contents.

(01 mark)

(b)Solutio

(i) Boiling can only remove temporary hardness of water because the temporary hardness is due to the presence of calcium or magnesium bicarbonates. ……01 mark

$Ca^{2+}{}_{(aq)} + 2HCO_3^-{}_{(aq)}$ …..00½ mark

Boiling promotes the formation of carbonate from the bicarbonate and precipitates calcium carbonate (or magnesium carbonate) out of solution, leaving water that is softer upon cooling (making the Ca^{2+} or Mg^{2+} unavailable to react with soap and cause the hardness): 01 mark

$Ca^{2+}{}_{(aq)} + 2HCO_3^-{}_{(aq)} \xrightarrow{Heat} CaCO_{3\,(s)} + CO_{2\,(aq)} + H_2O_{(l)}$ ……01 mark

However, it appears that the hardness of water in their case (Village B) is permanent; permanent hardness of water is due to the presence of calcium (or magnesium) sulphates and chlorides

$Ca^{2+}_{(aq)} + SO_4^{2-}_{(aq)}$ (01 mark)

or $Mg^{2+}_{(aq)} + + SO_4^{2-}_{(aq)}$ or chlorides ……00 ½

(ii) Permanent hardness of water can be removed by:

- Addition of Washing soda (sodium carbonate)

$Na_2CO_{3(aq)} + CaSO_{4(aq)} \rightarrow CaCO_{3(s)} + Na_2SO_{4(aq)}$ ……01 mark

The formed sodium salts are soluble, but do not cause the water to be hard. The calcium or magnesium ions are precipitated as insoluble calcium and magnesium carbonates. 01 mark

$Ca^{2+}_{(aq)} + CO_3^{2-}_{(aq)} \rightarrow CaCO_{3(s)}$

$Mg^{2+}_{(aq)} + CO_3^{2-}_{(aq)} \rightarrow MgCO_{3(s)}$ ……01 mark

8. (a) (i) Avogadro's Law states that equal volumes of all gases at the same temperature and pressure contain the same number of molecules. ……..……..01 mark

(ii) Solution

Change 5.6 dm^3 of ammonia to moles $=0.46dm^3/22.4dm^3=0.25$ mol. Then multiply by the Avogadro's constant to get the total number of molecules0: ……00½ mark

$25 \times 6.02 \times 10^{23} = 1.5 \times 10^{23}$ molecules ……...01½ marks

(b) Given:

Mass of $CaCO_3$ = 150 g

Mr of $CaCO_3$ = 40 + 12 +48 = 100 g/mol ……01 mark

$n_{CaCO3} = = \frac{150\ g}{100\ g/mol}$ = 1.5 moles ……01 mark

Reaction: $CaCO_3 \rightarrow CaO + CO_2$ ……00½ mark

For the reaction: 1 mol of $CaCO_3$ = I mol of CO_2

1.5 moles of $CaCO_3 \rightarrow$? (00½ mark)

$nCaCO_3$ = 1.5 moles……01 mark

Mass of CO_2 = n × Mr = 1.5 × 44

= 66 g ……01 mark

Mass left = Mass of $CaCO_3$ – Mass of CO_2

= 150 g – 66 g……01 mark

= 84 g……00½ mark

SECTION C (30 marks)

9. (a)

- Emulsion: a cloudy mixture of tiny droplets of one liquid suspended in another liquid. ……. 01½ mark
- Alloy: A mixture of two or more metals (or sometimes with a non-metal)…… 01½ mark
- Distillate: A liquid obtained after condensing the product of distillation…… 01½ mark
- Sublimation: Direct change from solid to gas…… 01½ mark

Solutions	Suspensions
…homogeneous	...heterogeneous …… 01½ mark
Transparent	Opaque …… 01½ mark
Particles completely dissolved	Particle separate on standing …… 01½ mark
Components separated by evaporation	Components separated by filtration…… 01½ mark

(b)

(i) All matter is made up of tiny discrete particles. …… ……01mark

…… 01mark

(ii) The particles are in constant motion.

(iii)The kinetic energy of a particle changes with temperature; the higher the temperature the faster the particle moves. 01mark

(Total 15 marks)

10.

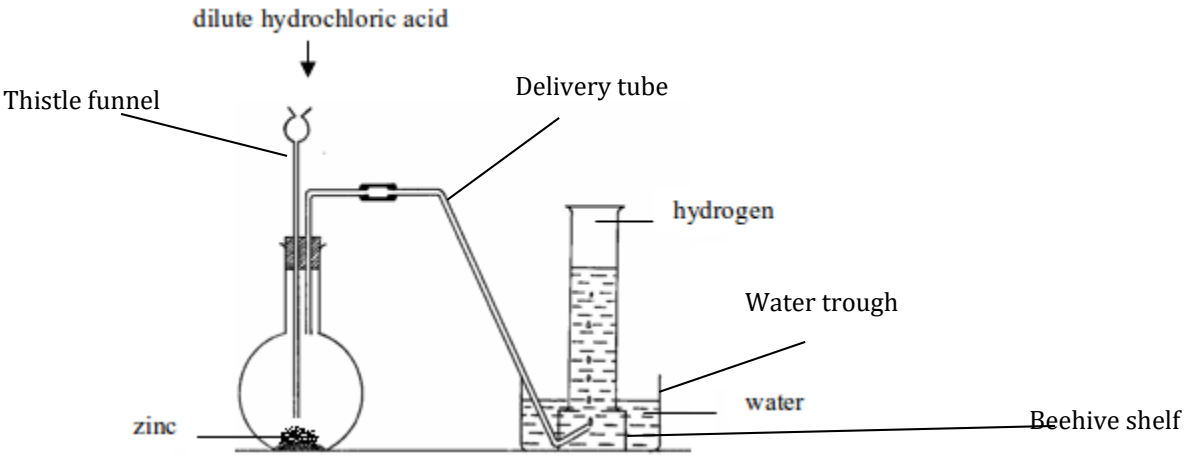


Diagram ……2 marks

Label (dilute Hydrochloric acid/sulphuric acid, zinc, hydrogen, water, water trough, delivery tube, funnel, beehive shelf…) any eight .....00½ @ = 04 marks

Properties of hydrogen

- It is a colourless, tasteless and odourless gas.
- It is almost insoluble in water (2 volumes of hydrogen gas dissolve in 100 volumes of water at 8ºC).
- It is the lightest of all gases. It is about 20 times lighter than air (one litre of hydrogen at 0ºC and 760 *mmHg* pressure weighs 0.0899 grams)
- It combines easily with other chemical substances at high temperatures. For example, it combines with oxygen to form water.
- Hydrogen acts as a reducing agent, by removing oxygen from some compounds. For example, copper (II) oxide is reduced to copper by heating it in a stream of hydrogen. The hydrogen is oxidized to water.
- $CuO_{(s)} + H_{2(g)} \rightarrow Cu_{(s)} + H_2O_{(l)}$
- It is neither acidic nor basic, so it a neutral gas .....01@ = 04 marks
- It does not support combustion.
- A mixture of hydrogen and oxygen explodes when lit.

Uses of Hydrogen Gas

- It is used in the manufacture of ammonia by the Haber process, which is based on the direct combination of hydrogen and nitrogen.
- It is used in the hardening of vegetable oils to make margarine.

- Hydrogen is used by meteorologists to fill weather balloons, which carry weather instruments that record information on various elements of weather in the upper atmosphere.
- It is as a fuel for space rockets in a form of water gas.When hydrogen contained in water gas is burned in air, it produces extremely high heat energy that is used to power rocket engines.
- It is used as an oxy-hydrogen flame in the welding and cutting of metals.
- It is used in the synthesis of hydrochloric acid. In this case, hydrogen combines directly with chlorine to form hydrogen chloride gas.
- The hydrogen chloride gas is then dissolved in water to form hydrochloric acid.
- It is used in the manufacture of methanol (wood spirit). In this process hydrogen combines directly with carbon monoxide.
- It is used in the reduction of metallic ores. .....01@ = 05 marks

(Total 15 marks)

11. The Importance of Changing One State of Matter to Another

- Separation of mixtures: It enables separation of mixtures can through such processes as distillation, sublimation, evaporation and condensation.
- Industrial manufacture of products: Industrially, the process of distillation is applied in the production of pure substances such as beer and other alcoholic drinks such as wine. The manufacturing process involves boiling, evaporating and condensation.
- Refining of petroleum (crude oil): Crude oil contains organic liquid components, each with a different boiling point. In the refinery, the components with lower boiling points evaporate first and get separated out, leaving those with higher boiling points behind. In this way, we get various types of oil components (fractions) such as petrol, diesel, kerosene, lubricating oil, etc.
- Drying of crops and clothes: When you suspend your clothing on a cloth line to dry, the moisture in it is lost through evaporation. Likewise, farmers in the village often spread crops on the ground to dry.
- Cooling of our bodies in hot weather: Cooling of our bodies in hot weather depends in the evaporation of sweat or water.
- Ice formation in refrigerators: You can freeze water into ice and then use the resulting ice for cooling the beverage or preserving foods.
- Melting metals to make alloys: In metallurgical industries, need may arise to mix two or more metals (alloys) together. This is only possible, where two or more metals are first melted at high temperatures into liquids. Then the resulting liquid metals are mixed in appropriate proportions. This is followed by cooling down the mixture to a solid alloy.
- Formation of rain: Rain is mainly formed through the process of evaporation and condensation. Water vapour, evaporating mostly from water bodies (oceans, seas, lakes, rivers, ponds, etc), land and plants rises up to the sky. As it rises, it cools down and condenses into tiny droplets

……Any six 02½ @ = 15 marks

(Total 15 marks)